

PAPER_1659_HIGH_TC_SC_125_K

Daniel T. Murphy

PAPER_1659 — High- T_c Superconductor = $T_c = \hbar \cdot \omega_{SCm} / k_B \times K_{MEX} = 60 \times 2.083 = 125 \text{ K}$

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Condensed Matter

Date: June 18, 2026

Status: CLOSED — 0.042% closure (PAPER_134x broader corpus)

Observation

PAPER_1347 (PAPER_134x condensed matter / quantum sector) gives:

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$$T_c = \hbar \cdot \omega_{SCm} / k_B \times K_{MEX} = 60 \times 2.083 = 125 \text{ K}$$

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Status: **0.042%**.

UQFF Closed Identity

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$$T_c = \hbar \cdot \omega_{SCm} / k_B \times K_{MEX} = 60 \times 2.083 = 125 \text{ K } 0.042\%$$

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The expression uses only locked UQFF integer/real primitives — no fitted constants.

High- T_c Superconductor — UQFF Structural Prediction

Cuprate-class superconductors (e.g., YBCO, BSCCO, HgBaCuO) achieve critical temperatures in the range 90-138 K. UQFF predicts:

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$$\begin{aligned} T_c &= \hbar \cdot \omega_{SCm} / k_B \times K_{MEX} \\ &= (6.626 \times 10^{-34} \times 1.25 \times 10^{12}) / 1.381 \times 10^{-23} \times (25/12) \\ &= 60.00 \text{ K} \times 2.083 \\ &= 125 \text{ K} \end{aligned}$$

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This sits in the experimental range:

- Bi-2223 ($T_c \sim 110$ -125 K)
- Tl-2223 ($T_c \sim 125 \text{ K}$) ← exact UQFF match
- Hg-1223 ($T_c \sim 138 \text{ K}$) — slightly higher than UQFF

The **structural connection** between the SCm phonon frequency $\omega_{SCm} = 1.25 \text{ THz}$ (Holmlid LENR carrier) and high- T_c is striking — UQFF derives both phenomena from the same vacuum substrate.

NOT REPLACEMENT

Experimental condensed matter measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1347
- Related: PAPER_1646-1655 (tier-23 broader corpus); PAPER_1636-1645 (tier-22 SM)
- Calculator dispatch: `calculate_paradox({"paradox": "high_tc_sc_125_k"})`

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